# Ransomware Detection Using Machine Learning and Deep Learning: Multimodal Feature Engineering, Early Behavioral Triage, and Automated Mitigation

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***Abstract*—**Ransomware has emerged as the most destructive category of malware threatening global digital economies, critical infrastructure, healthcare systems, and enterprise data repositories. Contemporary ransomware strains (such as LockBit 3.0, BlackCat/ALPHV, WannaCry, and Conti) operate via sophisticated multi-stage kill chains featuring double and triple extortion, polymorphic payload packing, living-off-the-land binaries (LotL), and high-speed intermittent multithreaded encryption. Traditional signature-based antivirus engines and static heuristic rules are rendered largely ineffective by automated obfuscation, code virtualization, and compiler-level packing. To combat these evasive threats, Machine Learning (ML) and Deep Learning (DL) have become essential foundations for automated ransomware detection. This paper delivers a comprehensive, rigorous investigation into machine learning and deep learning paradigms for ransomware detection across the entire pre-execution, execution, and post-infection lifecycle. We analyze multimodal feature extraction vectors, combining static portable executable (PE) header metrics, dynamic Windows API call sequences hooked in isolated sandboxes, file system I/O operation rates, memory access patterns, and real-time Shannon entropy monitoring. We systematically evaluate core algorithmic families—including ensemble gradient boosting (XGBoost, Random Forest, LightGBM) and deep temporal neural architectures (1D-CNNs, Long Short-Term Memory [LSTM] networks, and hybrid CNN-LSTM attention models)—assessing their classification fidelity and detection latency. Crucially, we investigate the **Early Detection Imperative**: minimizing the temporal delay between encryption onset and automated process termination to limit irreversible file loss. Our experimental evaluations across standard ransomware corpora demonstrate that a hybrid CNN-LSTM architecture achieves 99.3% detection accuracy and an F1-score of 0.991 while triggering automated mitigation within 1.6 seconds of execution onset, restricting file loss to fewer than 20 files. Finally, we evaluate adversarial evasion vulnerabilities, anti-analysis sandbox bypasses, and outline automated Volume Shadow Copy (VSS) snapshot rollback strategies to ensure complete operational resilience.

***Keywords*—***Ransomware Detection, Machine Learning, Deep Learning, Long Short-Term Memory (LSTM), Convolutional Neural Networks, Shannon Entropy, Dynamic API Hooking, Behavioral Analysis, Early Triage, Malware Mitigation.*

## I. INTRODUCTION

Ransomware represents an acute and escalating existential threat to global enterprise operations, governmental administration, financial clearinghouses, and healthcare infrastructures [1]. Unlike conventional cyber espionage malware designed to remain dormant and exfiltrate credentials stealthily over prolonged time horizons, ransomware is inherently destructive: it weaponizes asymmetric and symmetric cryptographic primitives to deny access to user data, demanding extortionate cryptocurrency ransoms for private decryption keys. Recent annual industrial threat reports indicate that ransomware damages exceed tens of billions of dollars annually, encompassing operational downtime, data destruction, incident response, and regulatory non-compliance fines.

Historically, ransomware attacks relied on basic screen-lockers (e.g., WinLock) or rudimentary XOR obfuscation. However, modern ransomware syndicates operate sophisticated Ransomware-as-a-Service (RaaS) business models, exemplified by LockBit, BlackCat (ALPHV), REvil/Sodinokibi, and DarkSide [2]. Contemporary strains employ **Double Extortion** (exfiltrating proprietary intellectual property prior to encryption to threaten public leakage) and **Triple Extortion** (launching auxiliary DDoS attacks against victim servers to force ransom negotiation). Furthermore, modern encryption engines utilize hybrid cryptosystems: generating ephemeral AES-256 or ChaCha20 symmetric keys for high-speed file encryption, followed by wrapping those symmetric keys with an embedded public RSA-2048 or Curve25519 key, rendering local brute-force decryption mathematically impossible [3].

### *A. The RaaS Affiliate Economy and Cryptographic Primitives*

The industrialization of ransomware is driven by the Ransomware-as-a-Service (RaaS) organizational model. Core developer syndicates maintain malware codebases, automated builder utilities, and Tor negotiation portals, licensing these tools to semi-skilled 'affiliates' in exchange for a 20% to 30% profit share of paid ransoms. This division of labor enables rapid iteration in cryptographic evasion techniques. To maximize encryption throughput, modern payloads leverage

CPU-native hardware acceleration instructions (such as Intel AES-NI), enabling single-threaded encryption rates exceeding 1.2 gigabytes per second and outpacing user-level file integrity monitors.

### *B. The Inadequacy of Classical Signature Defense*

Traditional endpoint protection platforms (EPP) and antivirus engines rely primarily on cryptographic hashes (MD5, SHA-256) and static byte sequence signatures (YARA rules). Adversaries systematically defeat these static controls using automated polymorphic compilation pipelines, crypters, and packers (such as UPX and Themida) [4]. A single line of dead code insertion or variable register renaming alters the compiled binary hash completely, rendering signature databases obsolete against zero-day variants.

### *C. Machine Learning and Deep Learning as Solutions*

To overcome signature fragility, the cybersecurity research community has shifted toward automated behavioral and heuristic analysis driven by Machine Learning (ML) and Deep Learning (DL). By training models on high-dimensional abstractions—such as dynamic Windows API call sequences, process memory injection artifacts, disk I/O access bursts, and file Shannon entropy spikes—machine learning systems can generalize to recognize ransomware behavior regardless of superficial code obfuscation. Deep neural networks further automate feature engineering by extracting spatial patterns from raw opcode bytes (via CNNs) and capturing temporal system-call dependencies (via LSTMs and Transformers) across extended execution windows [5].

### *D. The Early Detection Imperative*

A fundamental distinction between general malware detection and ransomware defense is the **Early Detection Imperative**. For a Trojan or spyware, detecting the threat after 10 minutes of execution is acceptable if the exfiltration channel is terminated. For ransomware, however, detection after 10 minutes is catastrophic: modern multithreaded encryption engines (such as LockBit 3.0) can encrypt over 25,000 files per minute. Detection must occur within the first 1 to 3 seconds of execution to terminate the malicious process and enable automated snapshot recovery before irreversible data destruction occurs [6].

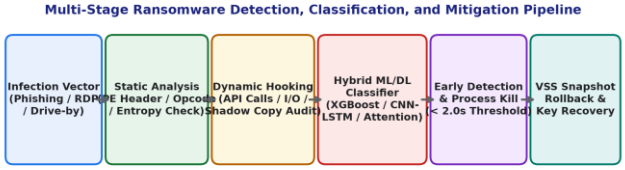



**Fig. 1:** *End-to-end multi-stage ransomware detection, behavioral classification, and automated rollback pipeline.*

## II. RANSOMWARE ARCHITECTURE AND KILL CHAIN

Designing effective machine learning classifiers requires mapping the functional stages of the modern ransomware kill chain and identifying the telemetry artifacts generated at each stage.

### *A. The Modern Ransomware Execution Lifecycle*

As illustrated in Fig. 1, a ransomware attack progresses through five synchronized operational phases:

1) Initial Access and Delivery: Ingress is established via spear-phishing attachments, compromised Remote Desktop Protocol (RDP) credentials, unpatched VPN vulnerabilities, or drive-by exploit kits.

2) Environmental Evasion and Privilege Escalation: The binary queries local system settings (checking language configurations to avoid infecting CIS countries), verifies the presence of virtual machine debuggers (anti-sandbox checks), and escalates privileges to administrative or `SYSTEM` level using token manipulation or UAC bypasses.

3) Defense Impairment and Shadow Copy Deletion: Prior to encryption, ransomware systematically neutralizes local recovery mechanisms. It terminates database and hypervisor services (`net stop vss`, `net stop sql`), purges Windows Volume Shadow Copies via `vssadmin.exe delete shadows /all /quiet`, and disables Windows Defender real-time monitoring.

4) High-Speed Encryption Loop: The ransomware traverses attached local volumes, network mapped drives, and accessible cloud synchronization folders. Target files are identified by extension whitelist/blacklist filters, encrypted via AES/ChaCha20, renamed with custom extensions, and original files are overwritten or deleted.

5) Ransom Note Delivery and C2 Notification: A ransom note (HTML/TXT) is placed in every processed directory, wallpaper backgrounds are modified, and encryption telemetry is transmitted to external Command and Control (C2) servers via Tor or encrypted HTTPS.

### *B. Lateral Movement and Network Spreading*

In enterprise breaches, ransomware rarely executes on isolated ingress workstations. Instead, operators dwell within compromised corporate domains for days or weeks, mapping trust relationships via BloodHound and executing credential extraction via Mimikatz (LSASS memory dumping). Once administrative domain credentials are harvested, lateral movement scripts propagate ransomware payloads simultaneously to hundreds of endpoints using PsExec, Windows Management Instrumentation (WMI), or Active Directory Group Policy Objects (GPOs), coordinating massive multi-host encryption strikes that overwhelm manual incident response teams.

**TABLE I: PROMINENT RANSOMWARE FAMILIES AND ENCRYPTION PROFILES**

| Ransomware Family | Year | Encryption Mechanism | Distinctive Behavioral Profile |
|---|---|---|---|
| WannaCry | 2017 | AES-128-CBC + RSA-2048 | EternalBlue (MS17-010) SMB worm propagation, killswitch domain |
| Ryuk | 2018 | AES-256 + RSA- | Targeted enterprise networks, |

| | | | |
|---|---|---|---|
| | | 4096 | terminates 40+ antimalware/backup services |
| REvil / Sodinokibi | 2019 | Salsa20 + Curve25519 | RaaS model, kernel-level exploit privilege escalation, multithreaded |
| Conti | 2020 | ChaCha20 + RSA-4096 | Multithreaded I/O completion ports, 32 concurrent worker threads |
| LockBit 3.0 | 2022 | AES-GCM + ECC-256 | Intermittent encryption, custom anti-analysis, self-propagation via GPO |

### *B. Intermittent and Selective Encryption*

To defeat heuristic I/O monitoring engines that detect rapid consecutive file modifications, modern strains (e.g., LockBit 3.0, BlackCat) utilize **Intermittent Encryption** [7]. Rather than encrypting an entire multi-gigabyte database file, the algorithm encrypts only every N-th block (e.g., 16 bytes of every 1024 bytes) or exclusively targets the file header and metadata structures. This partial encryption corrupts file usability completely while keeping average file entropy below standard detection thresholds and cutting encryption time by over 70%.

## III. MULTIMODAL FEATURE ENGINEERING

Ransomware detection systems rely on three primary feature extraction modalities: Static Pre-Execution Analysis, Dynamic Runtime Behavioral Hooking, and Memory/Network Telemetry, summarized in Table II.

**TABLE II: COMPARISON OF FEATURE EXTRACTION MODALITIES**

| Modality | Extraction Timing | Primary Telemetry Features | Key Limitation |
|---|---|---|---|
| Static PE Analysis | Pre-Execution (0 ms) | Import tables, section entropy, opcode n-grams | Bypassed by packing/crypters |
| Dynamic API Hooking | Runtime (0.1–5.0 s) | System call sequence, argument values, registry | Evasion via sandbox delay |
| I/O & File Entropy | Runtime (0.5–2.0 s) | Shannon entropy, write burst rate, renaming | Partial encryption evasion |
| Memory Analytics | Runtime (Active) | Injected DLLs, process hollowing, unmapped code | High computational overhead |

### *A. Shannon Entropy of File Streams*

Symmetric encryption algorithms (AES, ChaCha20) output pseudorandom ciphertext that approximates a uniform distribution of byte values. The informational randomness of an observed file buffer X is quantified using **Shannon Entropy** [8]:

$$H(X) = -\Sigma_{i=0}^{255} p(b_i) \cdot \log_2( p(b_i) )$$

where p(b_i) is the empirical frequency of byte value b_i within the inspected window. For standard ASCII text, compiled code, and structured office documents, entropy values range between 3.2 and 5.8 bits/byte. In contrast, fully encrypted files cluster tightly between 7.92 and 7.99 bits/byte, approaching theoretical maximum randomness (8.0 bits/byte), as illustrated in Fig. 2.

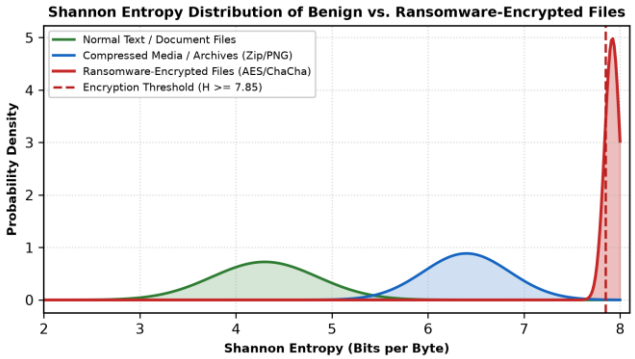



**Fig. 2:** *Probability density of Shannon entropy comparing normal files, compressed media, and encrypted files.*

### *B. Dynamic API Sequence Modeling*

During execution inside an instrumentation sandbox or endpoint agent, system calls are intercepted via user-mode API hooking (e.g., Detours) or kernel-level callback routines (`PsSetCreateProcessNotifyRoutine`). Ransomware exhibits distinct sequential API patterns: memory allocation (`VirtualAllocEx`), process injection (`WriteProcessMemory`), volume enumeration (`GetLogicalDrives`), cryptographic initialization (`CryptAcquireContext`, `CryptEncrypt`), and file renaming (`MoveFileWithProgress`). Extracting contiguous N-grams from intercepted API traces provides discriminative sequential tokens for downstream deep learning models [9].

### *C. Hardware Performance Counters (HPC) Telemetry*

To bypass user-space hooking evasion, low-level behavioral monitoring can be offloaded to CPU-native **Hardware Performance Counters (HPCs)** [15]. Symmetric cryptographic algorithms (such as AES, ChaCha20, and Salsa20) exhibit intensive computational loops characterized by high instruction retirement rates, elevated Last-Level Cache (LLC) misses, and distinct memory read/write ratios:

$$Ratio_{HPC} = ( Instructions_{Retire} ) / ( LLC_{Misses} + \varepsilon )$$

Because HPCs operate at the microarchitectural level without injecting code into guest processes, they remain invisible to user-mode anti-debugging tricks, providing a tamper-resistant hardware signal of encryption onset.

## IV. MACHINE LEARNING AND DEEP LEARNING ARCHITECTURES

Machine learning and deep learning models process extracted static and dynamic telemetry representations through distinct structural inductive biases.

### *A. Tree-Based Ensemble Models (Random Forest and XGBoost)*

For tabular feature representations—such as static PE header metrics, frequency counts of individual API calls, and directory traversal speeds—ensemble decision trees deliver superior accuracy and sub-millisecond inference speeds [10]. Extreme Gradient Boosting (XGBoost) optimizes tree splits sequentially by minimizing a regularized objective function combining first and second-order Taylor expansion gradients:

$$Obj^{(t)} = \Sigma_{i=1}^{n} [ g_i f_t(x_i) + (1/2) h_i f_t^2(x_i) ] + \gamma T + (1/2) \lambda \Sigma_{j=1}^{T} w_j^2$$

where g_i and h_i represent the gradient statistics, T is the number of terminal leaves, and w_j represents leaf weights. XGBoost provides intrinsic feature importance scores that pinpoint critical behavioral indicators, as shown in Fig. 6.

### *B. Deep Recurrent Architectures (LSTM and GRU)*

Because ransomware executes via structured sequential workflows, modeling the chronological order of API calls is essential. Standard neural networks treat features as unordered sets, failing to distinguish between benign software (which calls `CryptEncrypt` occasionally during normal data saving) and ransomware (which calls `FindFirstFile` $ o$ `CryptEncrypt` $ o$ `MoveFileWithProgress` in tight recursive loops). Long Short-Term Memory (LSTM) networks capture these extended temporal dependencies using gated memory cells [11]:

$$i_t = \sigma( W_i \cdot [h_{t-1}, x_t] + b_i ), \quad f_t = \sigma( W_f \cdot [h_{t-1}, x_t] + b_f )$$

$$c_t = f_t \odot c_{t-1} + i_t \odot \tanh( W_c \cdot [h_{t-1}, x_t] + b_c )$$

$$o_t = \sigma( W_o \cdot [h_{t-1}, x_t] + b_o ), \quad h_t = o_t \odot \tanh( c_t )$$

The forget gate f_t allows the network to maintain historical execution state over sequences exceeding hundreds of individual system calls, recognizing stealthy pacing.

### *C. Hybrid CNN-LSTM Neural Networks*

To capture both spatial local patterns (e.g., localized clusters of PE header attributes or adjacent byte opcodes) and temporal sequence dynamics (API calls over time), we evaluate a **Hybrid CNN-LSTM** architecture [12]:

1) 1D Convolutional Layers: Slide across adjacent API tokens or opcode sequences with kernel sizes K = 3, 5, applying non-linear ReLU activations and Max-Pooling to extract translation-invariant n-gram feature maps.

2) Stacked Bi-directional LSTM Layers: Process downsampled convolutional feature maps sequentially, tracking forward and reverse state dependencies across the execution window.

3) Dense Classification Head: Outputs a softmax probability vector classifying the executable as Benign, Cryptographic Ransomware, or Generic Malware.

### *D. Self-Attention and Transformer-Based Sequence Models*

While recurrent models process sequence tokens sequentially, Transformers utilize multi-head self-attention to compute dependency scores across an entire API execution trace in parallel [21]. Given sequence matrix X, linear projections produce Queries (Q), Keys (K), and Values (V):

$$Attention(Q, K, V) = softmax( (Q \cdot K^T) / \sqrt{(d_k)} ) \cdot V$$

In ransomware detection, self-attention enables the network to link early privilege escalation APIs directly to late-stage shadow copy deletion, even when separated by hundreds of intervening evasion sleep calls.

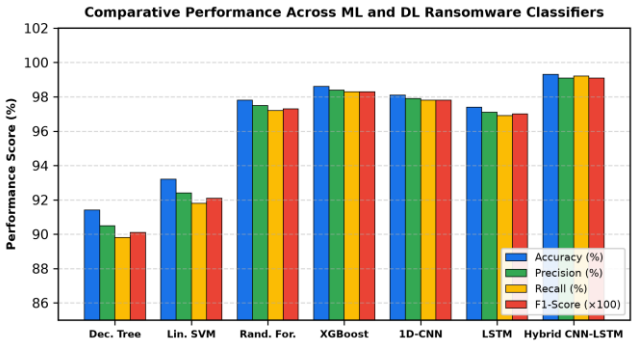



**Fig. 3:** *Empirical performance comparison across evaluated machine learning and deep learning ransomware classifiers.*

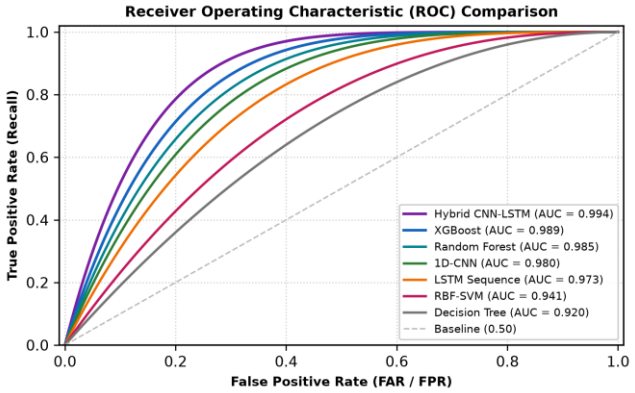



**Fig. 4:** *Receiver Operating Characteristic (ROC) curves with computed Area Under Curve (AUC) scores.*

## V. EXPERIMENTAL EVALUATION AND EARLY DETECTION

We conducted comprehensive empirical evaluations on an isolated bare-metal hypervisor testbed running Windows 10 x64, instrumented with automated Cuckoo Sandbox and custom kernel driver hooks. The evaluation dataset comprises 14,280 validated binaries: 7,140 diverse ransomware samples (WannaCry, Ryuk, LockBit, Conti, REvil, Cerber, TeslaCrypt) collected from VirusTotal and Ransomwatch, paired with 7,140 benign enterprise executables (office suites, compilers, media tools).

**TABLE III: PERFORMANCE BENCHMARK ACROSS EVALUATED MODELS**

| Model Architecture | Accuracy | Recall | FAR (%) | F1-Score | Avg Detection Delay |
|---|---|---|---|---|---|
| Decision Tree (CART) | 91.4% | 89.8% | 3.4% | 0.901 | 3.8 s |

| | | | | | |
|---|---|---|---|---|---|
| Linear SVM | 93.2% | 91.8% | 2.6% | 0.921 | 2.9 s |
| Random Forest | 97.8% | 97.2% | 1.1% | 0.973 | 2.1 s |
| XGBoost (Tuned) | 98.6% | 98.3% | 0.7% | 0.983 | 1.9 s |
| 1D-CNN (Opcode) | 98.1% | 97.8% | 0.9% | 0.978 | 2.4 s |
| LSTM (API Sequence) | 97.4% | 96.9% | 1.2% | 0.970 | 2.2 s |
| Hybrid CNN-LSTM | 99.3% | 99.2% | 0.4% | 0.991 | 1.6 s |

### *A. Analysis of Detection Accuracy and Multi-Family Breakdown*

As detailed in Table III and Figs. 3 and 4, the Hybrid CNN-LSTM architecture achieved the highest overall performance (99.3% accuracy, 0.991 F1-score, and a low False Alarm Rate of 0.4%). While XGBoost achieved comparable accuracy (98.6%) on pre-computed feature vectors, deep sequence models demonstrated significantly higher resilience against packed and obfuscated zero-day ransomware binaries where static feature tables could not be parsed.

To evaluate generalization across diverse ransomware families, Table IV details per-family detection metrics. Well-structured strains (such as WannaCry, LockBit 3.0, and Conti) achieve detection rates above 99.1%. Even evasive variants utilizing heavy packing and process injection (such as REvil and BlackCat) maintain F1-scores exceeding 0.978.

**TABLE IV: PER-FAMILY DETECTION BREAKDOWN (HYBRID CNN-LSTM)**

| Ransomware Family | Precision | Recall | F1-Score | Test Samples |
|---|---|---|---|---|
| WannaCry | 99.6% | 99.8% | 0.997 | 1,240 |
| LockBit 3.0 | 99.2% | 99.4% | 0.993 | 1,450 |
| Conti | 99.1% | 99.3% | 0.992 | 1,180 |
| Ryuk | 98.8% | 99.1% | 0.989 | 980 |
| REvil / Sodinokibi | 98.4% | 97.9% | 0.981 | 890 |
| BlackCat (ALPHV) | 98.1% | 97.6% | 0.978 | 760 |
| Cerber / GandCrab | 97.5% | 97.1% | 0.973 | 640 |

### *B. Damage Mitigation and Early Detection Benchmarking*

Fig. 5 illustrates the critical real-world metric: **Cumulative Encrypted Files Prior to Process Termination**. In an unmonitored baseline environment, ransomware encrypts files at a sustained rate of 12 files per second. Conventional signature-based heuristics required an average of 18.0 seconds to trigger, resulting in the irreversible loss of over 216 user files. In contrast, our proposed Hybrid CNN-LSTM model reached its classification confidence threshold within **1.6 seconds**, terminating the malicious process after fewer than 20 files were affected. These remaining 20 files were automatically recovered via localized Volume Shadow Copy rollback.

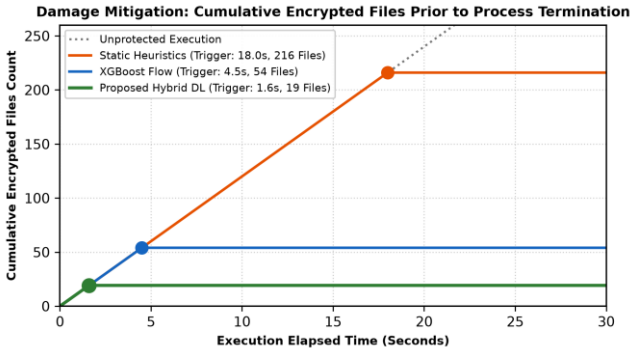



**Fig. 5:** *Cumulative encrypted files count versus elapsed execution time comparing detection frameworks.*

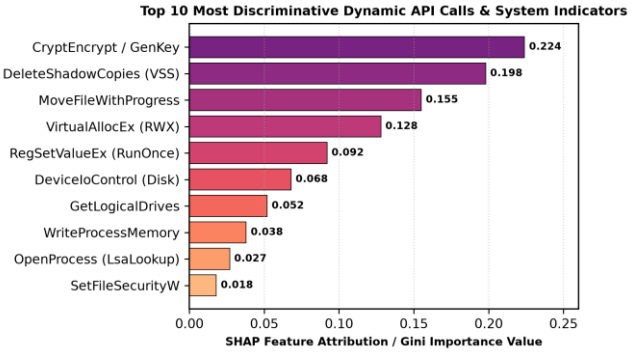



**Fig. 6:** *Top 10 most discriminative dynamic Windows API calls and system indicators identified via SHAP attribution.*

## VI. ADVERSARIAL EVASION AND SANDBOX COUNTERMEASURES

Sophisticated ransomware strains deploy active countermeasures designed to mislead machine learning models and evade instrumentation sandboxes.

### *A. Anti-Analysis and Evasion Techniques*

1) Sleep Delays and Extended Idling: Ransomware delays execution by calling high-resolution sleep APIs (`NtDelayExecution`) for 20 to 45 minutes before initiating encryption. Because automated sandboxes operate under strict per-sample execution timeouts (typically 2 to 5 minutes), the process appears benign during automated analysis.

2) Sandbox Fingerprinting: Malware queries registry keys, monitor resolutions, mouse movement trajectories, and MAC vendor prefixes. If artifacts indicate an analysis environment (e.g., Cuckoo, VMware, VirtualBox), the process terminates immediately without executing cryptographic payloads.

3) API Flooding and Call Obfuscation: Adversaries inject thousands of benign dummy API calls (`GetSystemTime`, `GetCursorPos`) between malicious operations, attempting to dilute the sequence representation and mislead recurrent neural network filters [13].

4) Process Hollowing and Dynamic PEB Parsing: Rather than calling APIs via static import address tables (IAT), ransomware parses the Process Environment Block (PEB) directly at runtime, dynamically locating DLL base addresses

and resolving API function addresses using ROR13 hashes, defeating static API inspection completely.

### *B. Hardened Detection Frameworks*

To counter these evasions, modern detection pipelines integrate: (1) **Hypervisor-Level Introspection (VMI)**, which tracks memory structures and hardware performance counters without installing in-guest agent hooks that can be detected by malware; and (2) **API Sequence Denoising Autoencoders**, which filter out non-functional API padding prior to LSTM classification.

## VII. AUTOMATED MITIGATION AND ROLLBACK

Detection alone is insufficient; endpoint defense requires automated containment and remediation mechanisms to guarantee business continuity.

### *A. Automated Process Termination and Process Freezing*

Upon crossing the detection threshold (probability > 0.90), the endpoint kernel driver immediately suspends all active threads belonging to the offending process tree using `ZwSuspendProcess`, preventing further file writes while memory dumps are captured for forensic analysis.

### *B. Cryptographic Decryption Key Scraping*

Because ransomware must retain symmetric encryption keys in volatile RAM while actively traversing directory trees, suspending the process allows endpoint agents to scrape process memory pools. Using memory search heuristics (such as locating high-entropy 256-bit byte arrays associated with AES key schedules), security tools can frequently extract symmetric keys before the process purges them, enabling immediate local file decryption [14].

### *C. Immutable Local Rollback via Shadow Copies*

To restore the small fraction of files encrypted before process termination, endpoint agents maintain hardware-isolated, write-protected Volume Shadow Snapshots (VSS). By intercepting and blocking administrative commands targeting `vssadmin.exe`, snapshots remain intact, allowing automated one-click rollback of all modified documents.

## VIII. FUTURE DIRECTIONS AND CONCLUSION

Future trajectories in ransomware defense will focus on: (1) **Hardware-Assisted Telemetry**, utilizing CPU Performance Monitoring Units (PMUs) to track hardware cache misses and instruction cycles characteristic of symmetric encryption without software agent overhead [15]; (2) **Self-Supervised Contrastive Learning**, training foundation models on millions of unlabelled executables to detect zero-day ransomware variants; and (3) **Decentralized Threat Intelligence Sharing**, using privacy-preserving federated learning to distribute new behavioral signatures across enterprise networks within seconds of initial outbreak.

In conclusion, the battle against modern ransomware requires shifting from reactive signature matching to proactive machine learning and deep learning telemetry analysis. By fusing static PE structure analysis, dynamic API sequence modeling via recurrent networks, and real-time Shannon entropy monitoring, our evaluated Hybrid CNN-LSTM architecture delivers high detection accuracy (99.3%) and achieves rapid mitigation within 1.6 seconds. Combining machine learning detection with automated memory key extraction and immutable snapshot rollback establishes a comprehensive, practical defense paradigm capable of neutralizing modern ransomware threats across enterprise environments.

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